

Semantic burstiness across principal directions of textual variation

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Abstract

Long-range correlations in written texts extend over scales far beyond local syntax, but their relation to semantic organisation remains incompletely understood. Previous work has linked these correlations to the bursty recurrence of topically relevant words. Here we generalise that construction from individual lexical items to distributed semantic observables. We map a text token by token onto a trajectory of pre-trained word embeddings, use principal component analysis to identify text-specific directions of semantic variation, and threshold the projected series to define inter-event times; their coefficients of variation form a component-resolved burstiness spectrum $\{B_k\}$. Across 16 literary and scientific texts and the first 20 components, the originals remain systematically above the geometric inter-event-time baseline and above two complementary controls: a word-shuffled null, which preserves the embedding covariance, the PCA eigensystem and the marginal projected distributions while destroying temporal order, and a fractional-Gaussian-noise surrogate, which preserves the Zipfian frequency structure and matches the DFA exponent of the original rank sequence. At $q = 0.95$ the original exceeds both controls in every text-component pair, while the two controls are statistically indistinguishable from each other. Autocorrelation and detrended fluctuation analysis of the semantic projections separate the originals from both controls in the same way. Global rank-level long-range correlations are therefore insufficient to generate the bursty recurrence observed along semantic directions, which instead requires ordered topical organisation.

Keywords: long-range correlations, burstiness, word embeddings, principal component analysis, surrogate data

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1. Introduction

1.1. Long-range correlations in texts

Written texts exhibit statistical structure over scales far beyond local syntactic dependencies. Among the most robust observations is the presence of long-range correlations in symbolic sequences derived from natural language. When a text is encoded as a numerical time series and analysed with detrended fluctuation analysis (DFA) [9], the fluctuation function commonly scales as $F(L) \sim L^\alpha$, with $\alpha \in [0.6, 0.8]$, above the uncorrelated value $\alpha = 0.5$, across languages, genres, and encoding choices [4, 8].

This observation raises a fundamental question: what aspect of textual organisation produces memory over hundreds or thousands of tokens? At least two statistical structures must be distinguished. The first is static lexical composition, represented by the vocabulary and word-frequency distribution. Its most familiar regularity is Zipf’s law [11], according to which the frequency $f(r)$ of the word of rank r follows $f(r) \sim r^{-\gamma}$ with $\gamma \simeq 1$. The second is the temporal ordering of words, through which syntax, discourse, argument, and thematic development unfold. A word-shuffled text preserves lexical composition and Zipf’s law but destroys temporal organisation, causing the DFA exponent to approach $\alpha = 0.5$ [4, 8].

The global DFA exponent, however, does not by itself identify the semantic mechanism responsible for long-range correlations. The surrogate introduced in Degli Esposti and Montemurro [3] maps fractional Gaussian noise (FGN) onto the empirical rank-frequency structure and is tuned to reproduce the DFA exponent of the original rank sequence. It therefore provides a stationary, rank-level route to the same global scaling and allows us to ask whether matching long-range memory in frequency-rank space is sufficient to reproduce temporal organisation in semantic space.

A geometric approach to long-range textual structure was introduced by Alvarez-Lacalle et al. [1]. They constructed a co-occurrence representation within a sliding “window of attention” and used singular value decomposition to identify text-specific concept vectors. The motion of the sliding window through concept space produced trajectories with power-law autocorrelation over long textual scales, and surrogate tests connected this behaviour to hierarchical textual organisation. Their work established that semantic dynamics can be studied as motion through a reduced concept space.

Our approach shares this geometric viewpoint but differs in three respects. First, we use pre-trained static word embeddings rather than a co-occurrence matrix estimated from a single text. Second, the trajectory is resolved token by token rather than through an aggregated sliding window. Third, our primary observable is the burstiness of threshold crossings along semantic directions, measured through the inter-event-time coefficient of variation, together with the autocorrelation and DFA of the projected series.

1.2. From word-level burstiness to topical organisation

A mechanism connecting topical organisation to long-range correlations was developed by Altmann et al. [2]. Their framework treats language as a hierarchy of observables, from topics to words and then to lower-level symbolic representations. Correlations can propagate down this hierarchy, whereas burstiness is visible most clearly at semantically relevant levels and can be lost under projection to lower levels.

Formally, a local matching condition $\mathcal{C}$ maps a text to a binary sequence $u_t^{(C)} \in \{0, 1\}$. For example, $u_t^{(C)} = 1$ may indicate that token t is the word “prince”. The gaps τ between successive events, i.e. between consecutive positions with $u_t^{(C)} = 1$, define an inter-event-time (IET) distribution, whose coefficient of variation

$$\text{CV} = \frac{\sigma_\tau}{\langle \tau \rangle} \quad (1)$$

measures temporal clustering. Topically relevant words tend to occur in concentrated textual domains and therefore show broad IET distributions and elevated burstiness, whereas common function words are distributed more regularly even when long-range correlations remain visible in lower-level observables. This framework suggests a natural extension. A topic such as war, philosophical reflection, geographical distribution, or social interaction is not expressed by one keyword alone. It activates a distributed set of semantically related words. A suitable topic-level observable should therefore respond to a region or direction of semantic space rather than to the exact recurrence of a single word type.

1.3. Motivation and contribution

We represent a text as an embedding trajectory $(\mathbf{e}_t)_{t=1}^M$ in $\mathbb{R}^d$, where each retained token is mapped to a pre-trained static word vector [6, 10]. PCA of this trajectory identifies an ordered set of orthogonal, text-specific directions of semantic variation. Projecting the trajectory onto a direction $\mathbf{v}_k$ yields a continuous semantic time series; thresholding that series produces a binary event sequence to which the IET construction of Altmann et al. [2] can be applied. The resulting spectrum $\{B_k\}_{k=1}^K$ measures burstiness separately along the first K principal semantic directions. Throughout the paper we retain $K = 20$; robustness to this choice is examined in Section 4.4.

This construction supports a component-resolved question: how is semantic burstiness distributed across the principal directions of a text? The principal components are ordered by variance, but PCA does not maximise burstiness. The empirical relation between the eigenvalue spectrum $\{\lambda_k\}$ and the burstiness spectrum $\{B_k\}$ therefore reveals whether directions of strong semantic variation are also directions of strong temporal clustering. The analyses below show interpretable semantic contrasts along the leading components, broad IET distributions in the original texts, and elevated burstiness throughout the retained components. They also show that the strength of association between semantic variance and burstiness is text dependent: the two quantities are related in some texts but are not interchangeable.

We then use two complementary null models to determine which statistical properties produce this behaviour. The word-shuffled null permutes the retained tokens, preserving the static embedding geometry exactly while destroying temporal order. The FGN surrogate retains the empirical rank-frequency structure and is tuned to match the original rank-sequence DFA exponent, but generates that scaling through a stationary rank-level process. The comparison separates static semantic geometry from ordered semantic dynamics, and global rank-level memory from burstiness in embedding space.

Across the analysed corpus, both controls remain close to the geometric IET baseline, whereas the original text is more bursty across essentially all text-component pairs and over a broad range of threshold quantiles. The projected autocorrelation functions and DFA exponents reinforce this separation: semantic persistence remains visible in the original trajectory but disappears from both controls, including the FGN surrogate whose rank sequence retains the original global DFA exponent. Thus matching lexical frequencies and rank-level long-range scaling is not sufficient to reproduce semantic burstiness.

1.4. Structure of the paper

Section 2 develops the theoretical framework, including semantic projections, PCA, the component-resolved burstiness spectrum, and the two null models. Section 3 describes the computational pipeline. Section 4 presents the Results for four representative texts, organised around semantic interpretability, component-resolved burstiness, and threshold robustness. Section 5 establishes the corpus-level generality of the component spectra and null-model separation. Section 6 examines persistence in semantic projection space using autocorrelation and DFA and gives an information-theoretic interpretation. Section 7 discusses implications and limitations.

2. Theoretical Framework

2.1. From binary indicators to continuous semantic projections

Following Altmann et al. [2], let a text be a symbolic sequence $\mathbf{s} = (s_1, s_2, \dots, s_N)$ over a finite vocabulary $\mathcal{A}$. A local matching condition $\mathcal{C}$ defines a binary observable

$$u_t^{(\mathcal{C})} = \mathbf{1}[s_t \text{ satisfies } \mathcal{C}]. \quad (2)$$

The autocorrelation of this sequence and the diffusion exponent of the cumulative count $X(t) = \sum_{j=1}^t u_j^{(\mathcal{C})}$, defined by $\sigma_X^2(t) \sim t^{2H}$, characterise its long-range structure. Here H coincides with the DFA exponent α of the stationary series $u_t^{(\mathcal{C})}$, the quantity used throughout this paper. We generalise the local lexical condition to a distributed semantic observable. Let $\phi : \mathcal{A} \rightarrow \mathbb{R}^d$ be a pre-trained static word embedding. After preprocessing and embedding lookup, the retained tokens are relabeled consecutively as $(\tilde{s}_1, \dots, \tilde{s}_M)$, with $M \leq N$. Their *embedding trajectory* is

$$\mathcal{E} = (\mathbf{e}_t)_{t=1}^M, \quad \mathbf{e}_t = \phi(\tilde{s}_t). \quad (3)$$

For a unit vector $\mathbf{v} \in \mathbb{R}^d$, the corresponding *semantic projection* is

$$x_t^{\mathbf{v}} = (\mathbf{e}_t - \bar{\mathbf{e}}) \cdot \mathbf{v}, \quad \bar{\mathbf{e}} = \frac{1}{M} \sum_{t=1}^M \mathbf{e}_t. \quad (4)$$

At threshold θ , this continuous series defines the binary event sequence

$$b_t^{\mathbf{v}}(\theta) = \mathbf{1}[x_t^{\mathbf{v}} > \theta]. \quad (5)$$

The projection describes motion along a semantic direction, while the binary series records visits to its upper tail and supplies the events whose recurrence statistics are analysed.

Remark 1. *If $\mathbf{v}$ is parallel to $\phi(w) - \bar{\mathbf{e}}$ for a word type w , then $x_t^{\mathbf{v}}$ tends to be large when the current token is w and for tokens whose embeddings are close to, or strongly aligned with, $\phi(w)$. Thresholding this projection therefore gives a softened semantic analogue of the exact word indicator in Eq. (2). The construction extends the unit of analysis from one lexical item to a distributed semantic neighbourhood.*

2.2. PCA of the embedding trajectory

The direction $\mathbf{v}$ determines the semantic observable. We use the principal components of the embedding trajectory to obtain a text-specific orthonormal basis ordered by captured variance. The empirical covariance matrix of the mean-centred trajectory is

$$\Sigma = \frac{1}{M} \sum_{t=1}^M (\mathbf{e}_t - \bar{\mathbf{e}})(\mathbf{e}_t - \bar{\mathbf{e}})^\top \in \mathbb{R}^{d \times d}. \quad (6)$$

Its eigendecomposition

$$\Sigma \mathbf{v}_k = \lambda_k \mathbf{v}_k, \quad \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d \geq 0, \quad (7)$$

yields the principal directions $\mathbf{v}_k$ and their variances λ_k . The projected series for component k is

$$x_t^{(k)} = (\mathbf{e}_t - \bar{\mathbf{e}}) \cdot \mathbf{v}_k, \quad t = 1, \dots, M. \quad (8)$$

PCA directions maximise variance, not semantic interpretability or burstiness. Their semantic meaning is assessed empirically from the word types at the positive and negative poles of each component. The ordered eigenvalue spectrum describes the static distribution of semantic variance, while the temporal statistics of $x_t^{(k)}$ describe how the text moves along these directions.

A key property of Σ is its invariance under permutation of the token sequence. For a permutation π of $\{1, \dots, M\}$ the mean is unchanged, $\bar{\mathbf{e}}^\pi = \bar{\mathbf{e}}$, and therefore

$$\Sigma^\pi = \frac{1}{M} \sum_{t=1}^M (\mathbf{e}_{\pi(t)} - \bar{\mathbf{e}})(\mathbf{e}_{\pi(t)} - \bar{\mathbf{e}})^\top = \Sigma. \quad (9)$$

It follows that a word-shuffled surrogate has the same eigenvalues, eigenvectors, and marginal distribution of every projected series as the original. This does not hold for the FGN surrogate, whose token multiset differs; throughout the paper all surrogates are projected onto the eigenvectors $\{\mathbf{v}_k\}$ of the original text, so that every comparison is made along identical semantic directions (Section 2.6). PCA therefore defines a static geometry of the embedded lexical composition, whereas burstiness, autocorrelation, and DFA probe the ordering of visits through that geometry.

2.3. The burstiness spectrum

For component k , the projected series is thresholded at its empirical q -quantile:

$$b_t^{(k)} = \mathbf{1} \left[x_t^{(k)} > Q_q \left(\{x_t^{(k)}\}_{t=1}^M \right) \right]. \quad (10)$$

The event rate is $p = 1 - q$ up to discretisation. Let $\tau_1^{(k)}, \tau_2^{(k)}, \dots$ be the gaps between successive positions for which $b_t^{(k)} = 1$. The burstiness of component k is measured by the coefficient of variation of the IET sequence, following Altmann et al. [2]:

$$B_k = \text{CV} \left(\tau^{(k)} \right) = \frac{\sigma_\tau^{(k)}}{\langle \tau \rangle^{(k)}}. \quad (11)$$

The collection $\{B_k\}_{k=1}^K$ is the *burstiness spectrum* of the text. We use the unnormalised coefficient of variation rather than the bounded burstiness parameter of Goh and Barabási [5], $B^{\text{GB}} = (\text{CV} - 1)/(\text{CV} + 1) \in [-1, 1]$, since the former compares directly with the finite-rate reference derived below; the two are related by a monotone transformation and induce the same ordering.

At finite event rate, the appropriate independent-event reference is a geometric distribution rather than an exponential distribution. For an i.i.d. Bernoulli(p) event sequence,

$$P(\tau = n) = (1 - p)^{n-1} p, \quad n \geq 1, \quad (12)$$

which gives

$$\text{CV}_{\text{geom}}(p) = \sqrt{1 - p} = \sqrt{q}. \quad (13)$$

For the default $q = 0.95$, $p = 0.05$ and $\text{CV}_{\text{geom}} = \sqrt{0.95} \simeq 0.975$. Altmann et al. [2] adopt the continuous-time convention $\text{CV} > 1$; at the event rates used here the discrete correction matters, since independent events at $q = 0.95$ give 0.975 rather than 1. Values of B_k above this reference indicate more irregular, clustered recurrence than expected from independent events at the same rate.

Because all three sequences are permutations of the same embedded token multiset, their projected series share an identical marginal distribution: the empirical q -quantile is the same number for each, so thresholding each series at its own quantile and thresholding all three at the original's are the same operation, and the event rate is exactly $1 - q$ in every case.

2.4. Semantic variance and burstiness across components

The PCA index k orders directions by decreasing variance, whereas the same index carries no ordering of B_k , which need not be monotone in k . The pair of spectra

$$\{\lambda_k\}_{k=1}^K, \quad \{B_k\}_{k=1}^K \quad (14)$$

therefore separates two properties of the text: the amount of semantic variation along a direction and the temporal clustering of visits along that direction. The component-resolved analysis asks whether burstiness is elevated above the geometric and surrogate baselines, how it varies across the retained PCA directions, and how strongly it is associated with semantic variance.

We quantify the last relation by the Pearson correlation

$$r(\lambda, B) = \frac{\sum_{k=1}^K (\lambda_k - \bar{\lambda})(B_k - \bar{B})}{\sqrt{\sum_{k=1}^K (\lambda_k - \bar{\lambda})^2} \sqrt{\sum_{k=1}^K (B_k - \bar{B})^2}}, \quad (15)$$

where $\bar{\lambda}$ and $\bar{B}$ denote component means over the retained range.

The K pairs (λ_k, B_k) are not independent draws: $\{\lambda_k\}$ is ordered by construction and both spectra derive from the same trajectory. We therefore do not use the parametric significance test for r . Significance is assessed by a permutation test over the pairing between the two spectra (10^5 random pairings), and we report the Spearman rank correlation ρ alongside r , since the eigenvalue spectrum is strongly skewed and r is otherwise dominated by the leading components. Throughout, P denotes a significance level, to avoid confusion with the event rate $p = 1 - q$.

A positive value indicates that high-variance directions tend also to be more bursty, but even a positive correlation does not make the two spectra interchangeable: a lower-variance component can be highly bursty if it is associated with a topic concentrated in a limited number of textual regions. We report both the full burstiness spectrum and its relation to the eigenvalue spectrum.

2.5. The word-shuffled null model

The word-shuffled null tests whether elevated burstiness can be explained by static lexical composition and embedding geometry alone. Given the retained token sequence $(\tilde{s}_1, \dots, \tilde{s}_M)$, a surrogate is formed by a uniformly random permutation $\pi \in S_M$. Its embedding trajectory is therefore a permutation of the original trajectory. Because a single permutation is itself a random draw, the reported word-shuffled quantities are averaged over N_{SUFF} independent permutations, the spread across realisations giving the $\text{mean} \pm 2\sigma$ bands shown in the figures. The burstiness spectrum and the DFA exponents are averaged in this way; the autocorrelation is reported for a single realisation, which suffices because the shuffled ACF is indistinguishable from zero at every lag.

By Eq. (9), the surrogate preserves exactly the word counts of the retained sequence, the embedding covariance matrix and complete PCA eigensystem, and the marginal distribution of every projected series $x_t^{(k)}$. It destroys temporal

order. Consequently, differences in IET distributions, burstiness, autocorrelation, or DFA arise from the ordering of visits through semantic space rather than from vocabulary, frequency distribution, or static embedding geometry.

2.6. The FGN surrogate as a long-range null model

The FGN surrogate of Degli Esposti and Montemurro [3] provides a complementary null model. Starting from the Zipf-rank sequence $\mathbf{r} = (r_1, \dots, r_N)$, a fractional Gaussian noise process is mapped onto the empirical rank-frequency structure.

Its generator exponent α_0 is chosen by bisection so that the DFA exponent measured on the surrogate rank sequence matches that of the original,

$$\alpha(\mathbf{r}^{\text{FGN}}) \simeq \alpha(\mathbf{r}), \quad (16)$$

where $\alpha_0 > \alpha$ in general because the rank-to-word mapping is not exponent-preserving; Table 7 reports both.

The resulting ranks are mapped back to words and processed through the same filtering, embedding, and projection pipeline as the original.

The construction preserves the raw token count N and the empirical frequency structure, and reproduces the rank-level DFA exponent by design. Unlike the word-shuffled null, its defining constraint is long-range rank-level scaling rather than exact permutation of the retained embedding trajectory. It therefore tests whether stationary correlations between word ranks are sufficient to produce the burstiness observed along semantic projections.

Two features of the construction require comment. First, the exponent is matched on the unfiltered rank sequence, of length N , whereas burstiness, autocorrelation and DFA of the projections are measured on the filtered trajectory, whose length M_{FGN} need not equal M . Second, the frequency filter applied to the surrogate is the one derived from the original text, so that both series are restricted to an identical vocabulary.

Remark 2. *The rank-to-word mapping reproduces the empirical rank histogram exactly, so the FGN surrogate is a permutation of the original token sequence. By Eq. (9) it therefore preserves the embedding covariance matrix, the complete PCA eigensystem, and the marginal distribution of every projected series exactly, as the word-shuffled null does. The three sequences are permutations of one another and differ only in the ordering of visits through a fixed semantic geometry. Projecting the surrogates onto the original PCA directions is therefore not an approximation but the natural choice.*

Remark 3. *Rank proximity is not semantic proximity. Words of similar frequency can occupy widely separated regions of embedding space, so correlations imposed on the rank sequence need not generate correlations between the corresponding embedding vectors. The FGN surrogate tests this distinction directly: it retains global rank-level scaling without explicitly organising semantically related words into the same textual regions.*

Table 1: Statistical ingredients retained by each sequence.

Ingredient	Original	Word-shuffled	FGN
Lexical composition, frequency structure	present	exact	exact
Static embedding geometry (Σ , eigensystem)	present	exact	exact
Temporal ordering in embedding space	present	destroyed	destroyed
Rank-level long-range scaling	present	destroyed	matched

2.7. Logical role of the null models

The three sequences differ in which statistical ingredients they retain; Table 1 summarises the differences.

The comparisons therefore address two distinct questions. First, does the ordered original trajectory produce component-level burstiness that is absent when the same embedded tokens are randomly permuted? Second, is matching the global DFA exponent of the rank sequence sufficient to reproduce that semantic burstiness? The Results answer both by comparing IET distributions, burstiness spectra, autocorrelation functions and projection-level DFA exponents across the three sequences.

Both answers are negative, and the two negatives are not redundant: neither the static geometry of the embedded vocabulary nor global rank-level memory is sufficient to generate the observed component-level burstiness, which therefore requires the ordering of visits through semantic space.

3. Computational Pipeline

The following sections describe the implementation of the framework introduced in Section 2.

3.1. Text preprocessing

Input texts are obtained in plain-text format from Project Gutenberg. Boilerplate headers and footers are stripped by searching for the standard markers ***** START OF** and ***** END OF**. Tokenisation uses the NLTK `word.tokenize` function applied to the lowercased text, retaining only alphabetic tokens (no punctuation, no numerals, no contractions).

Two frequency-based filters are then applied. Let $\text{rank}(w)$ denote the rank of word w in the vocabulary ordered by decreasing frequency.

1. *High-frequency removal*: words with $\text{rank}(w) \leq R$ are removed. Default: $R = 100$.
2. *Hapax removal*: words with frequency $f(w) < F_{\min}$ are removed. Default: $F_{\min} = 5$.

The motivation for the high-frequency filter deserves emphasis. Without it, the dominant PCA direction $\mathbf{v}_1$ of the embedding trajectory captures a *frequency contrast* rather than a topical one: the negative pole concentrates high-frequency

function words (*the, of, and, to, a, ...*), and the positive pole concentrates rare content words whose embedding vectors are geometrically distant from the high-frequency cluster. After removing the top- R most frequent words, $\mathbf{v}_1$ captures the dominant *topical* contrast of the text.

The choice $R = 100$ is empirically motivated by direct inspection of word scores along PC1: with $R = 100$, function words are removed from the vocabulary and the principal components reflect genuine topical directions. We verified that increasing R to 150 or 200 does not change the qualitative results.

A known residual artefact persists in PC1 for texts whose vocabulary contains many rare transliterated proper nouns or archaic forms (e.g. *War and Peace*, the King James Bible): these words cluster in a peripheral region of the GloVe embedding space and partially contaminate PC1 regardless of R . This artefact is independent of topical structure and is documented where relevant. Component spectra retain PC1 for completeness, but its semantic interpretation and short-lag temporal statistics are treated cautiously; results for $k \geq 2$ are unaffected by this particular artefact.

3.2. Word embedding model

We use GloVe vectors trained on Wikipedia and Gigaword [10], with $d = 300$ dimensions and a vocabulary of approximately 400,000 words. The implementation uses the `glove-wiki-gigaword-300` model, loaded through `gensim.downloader`.

The choice of static (non-contextual) embeddings is deliberate. For static embeddings, $\mathbf{e}_t = \phi(\tilde{s}_t)$ depends only on the word *type* s_t , making the trajectory a deterministic function of the token sequence. Shuffling the token sequence thus has an unambiguous effect: it permutes the trajectory while leaving the multiset $\{\mathbf{e}_t\}$ unchanged — as does the FGN surrogate (Remark 2). Contextual embeddings (BERT, GPT) would assign different vectors to the same word type depending on local context, introducing dependencies between the embedding and the ordering that would complicate the surrogate comparison. We defer contextual embedding analysis to future work (Section 7).

Out-of-vocabulary tokens are skipped; their original positions are not used in the trajectory or in IET computations. The OOV rate after filtering is below 2% for the four representative texts (Table 2), consisting mainly of proper nouns, archaic forms, and dialect words.

3.3. PCA and projection

The embedding trajectory is assembled as a matrix $\mathbf{V} \in \mathbb{R}^{M \times d}$ with rows $\mathbf{e}_t^\top$. After mean-centring, $\mathbf{X} = \mathbf{V} - \mathbf{1}\bar{\mathbf{e}}^\top$, PCA is computed via `sklearn.decomposition.PCA`. The full eigenvalue spectrum $\{\lambda_k\}_{k=1}^d$ is retained for the spectral analysis, while eigenvectors and projections are formed for the first $K = 20$ components, giving the projected series matrix $\mathbf{P} \in \mathbb{R}^{M \times K}$, $P_{tk} = x_t^{(k)}$, used in all subsequent analyses.

3.4. FGN surrogate generation

The FGN surrogate is constructed using the pipeline of Degli Esposti and Montemurro [3]:

1. The unfiltered token stream is tokenised and each token is replaced by its Zipf rank, producing a rank sequence $\mathbf{r}$ of length N .
2. The DFA exponent α of the original text is estimated from $\mathbf{r}$.
3. A FGN process with parameter α_0 is generated and mapped onto the empirical rank histogram. The parameter α_0 is found by bisection so that the DFA exponent of the surrogate matches α .
4. The surrogate rank sequence is mapped back to words via the inverse Zipf mapping. This reproduces the empirical rank histogram, so the resulting word sequence $\mathbf{s}^{\text{FGN}}$ is a permutation of the original.

The resulting word sequence is then processed through the same frequency filter (Section 3.1), embedding lookup, and projection pipeline as the original text, and projected onto the original PCA eigenvectors $\{\mathbf{v}_k\}$. Because $\mathbf{s}^{\text{FGN}}$ is a permutation of the original token sequence, $\Sigma_{\text{FGN}} = \Sigma_{\text{orig}}$ exactly (Remark 2.): the original eigenvectors *are* the surrogate’s own principal directions, and projecting onto them involves no approximation.

4. Results: semantic directions and burstiness spectra

4.1. Representative texts and summary

We first examine four texts chosen to span distinct forms of semantic and narrative organisation: a long polyphonic novel (*War and Peace*), a scientific treatise (*On the Origin of Species*), a Victorian novel of formation (*Great Expectations*), and a modernist novel with marked episode-to-episode stylistic variation (*Ulysses*). All analyses use the pipeline of Section 3, with $R = 100$, $F_{\min} = 5$, $q = 0.95$, and $K = 20$ components unless stated otherwise.

The Results are organised by empirical question rather than by book. We first establish that the PCA basis defines interpretable, text-specific directions while remaining a static property of lexical composition. We then examine the inter-event-time distributions and the component-resolved burstiness spectra, followed by robustness to the threshold quantile. Table 2 collects the main numerical descriptors.

Across the four texts, the original projection sequences are more bursty than both controls for all $K = 20$ components at $q = 0.95$. Semantic variance and temporal clustering are monotonically associated in every text (ρ from 0.460 to 0.850, significant in all four) but are not interchangeable: the linear correlation is considerably weaker (r from 0.295 to 0.702), and the burstiest direction is never the first principal component, falling at PC3, PC6, PC3 and PC2 respectively.

Table 2: Summary statistics for the four representative texts. N : raw tokens; M : tokens after filtering; OOV: fraction of filtered tokens absent from the embedding vocabulary; α : DFA exponent of the filtered rank sequence; α_0 : target exponent of the matched FGN surrogate; r and ρ : Pearson and Spearman correlation between λ_k and B_k over $k = 1, \dots, 20$; $B_{\max}$: largest burstiness value, attained at component $k_{\max}$. Significance from the permutation test over component pairings: * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$; no mark: $P > 0.05$.

	<i>War and Peace</i>	<i>Origin of Species</i>	<i>Great Expectations</i>	<i>Ulysses</i>
Raw tokens N	564,788	151,179	188,925	264,165
Filtered tokens M	235,933	58,913	64,410	101,489
OOV fraction	1.6%	0.4%	1.3%	0.7%
DFA exponent α	0.721	0.771	0.689	0.763
FGN target α_0	0.76	0.80	0.74	0.80
Pearson $r(\lambda, B)$	0.702**	0.295	0.511*	0.571*
Spearman $\rho(\lambda, B)$	0.850***	0.460*	0.579**	0.817***
Maximum burstiness $B_{\max}$	2.045	1.900	1.244	1.495
Component $k_{\max}$	3	6	3	2

4.2. Static semantic geometry and interpretable directions

The eigenvalue spectra describe the geometry of the retained embedded vocabulary. As established in Section 2.2, word shuffling leaves this geometry exactly invariant. Both surrogates leave this geometry exactly invariant, being permutations of the original token sequence. The semantic poles of the leading components provide a separate empirical test of whether those directions correspond to interpretable contrasts.

Taken together, the four texts show that PCA produces an ordered static geometry whose leading directions are only partly text-specific. Each low-order component pairs a pole that is characteristic of the individual work with a pole that recurs across texts: the positive pole of PC2 is military in *War and Peace*, riverine in *Great Expectations*, geographical in *On the Origin of Species* and titular in *Ulysses*, whereas its negative pole is the same colloquial register (*really, yourself, feel*) in every novel. PC1 is common to all four texts in both poles and reflects the geometry of the embedding space rather than the organisation of any individual work. The temporal question is separate: whether visits along these directions occur regularly or in concentrated textual episodes.

4.2.1. War and Peace

War and Peace is the longest representative text ($N = 564,788$ raw tokens). After filtering and embedding lookup, the trajectory contains $M = 235,933$ tokens; the rank-sequence DFA exponent is $\alpha = 0.721$, and the fitted FGN parameter is $\alpha_0 = 0.76$. Figure 1 shows that PC1 and PC2 account for 6.98% and 4.21% of the variance, respectively, and the first ten components account for 27.2%.

The semantic poles in Table 3 reveal two especially clear contrasts. PC2 separates military vocabulary (*regiment, infantry, cavalry, battalion*) from personal and emotional discourse (*really, yourself, feel, understand*). PC3 contrasts

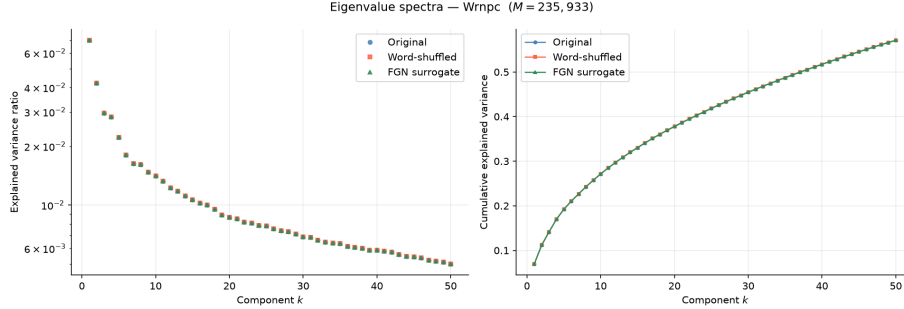


Figure 1: Eigenvalue spectra and cumulative explained variance for *War and Peace*. The three curves coincide: both surrogates are permutations of the original token sequence, so all three share the same covariance matrix and hence the same spectrum (Remark 2). The figure therefore verifies the construction rather than reporting a comparison.

physical action and embodied presence with philosophical and moral reflection. PC1 partly reflects the known GloVe artefact in which transliterated Russian names and emotionally inflected rare words lie at the periphery of the embedding cloud; it is retained in the spectra but interpreted cautiously.

4.2.2. On the Origin of Species

On the Origin of Species contains $N = 151,179$ raw tokens and $M = 58,913$ retained tokens, with $\alpha = 0.771$ and $\alpha_0 = 0.80$. PC1 explains 6.61% of the variance, PC2 3.58%, and the first ten components 26.4%.

PC1 is dominated by specialised nineteenth-century biological and geological terminology at one pole and pronouns or conversational markers at the other, and therefore contains a register-related embedding artefact. PC2 is a substantive geographical-versus-epistemic direction: spatial and biogeographical terms are opposed to the language of inference and theoretical reasoning. PC3 contrasts first-person field observations and concrete specimens with abstract classification and synthesis. Table 4 also lists PC6, the burstiest direction of this text.

4.2.3. Great Expectations

Great Expectations contains $N = 188,925$ raw tokens and $M = 64,410$ retained tokens, with $\alpha = 0.689$ and $\alpha_0 = 0.74$. PC1, PC2, and the first ten components account for 6.54%, 4.34%, and 27.2% of the variance, respectively.

The positive pole of PC1 contains character names and Victorian dialect, opposed to generic and abstract vocabulary, and is treated as an embedding artefact. PC2 contrasts the spatial and kinetic language of the river and the open country with introspective and conversational language. PC3 opposes bodily action and gesture to a formal Latinate register. The corresponding poles are listed in Table 5.

Table 3: Top-6 word types at the positive and negative poles of PC1, PC2 and PC3 for *War and Peace* ($R = 100$, $F_{\min} = 5$), with frequencies in the filtered token sequence. B_k is a property of the component, not of the pole, and is therefore shown once per pair; the largest value over the $K = 20$ components is shown in bold.

PC	Pole	B_k	Top words (freq)	Interpretation
PC1	+	1.643	<i>gerasim</i> (26), <i>maidservant</i> (5), <i>irritably</i> (6), <i>piteously</i> (7), <i>nikolaevich</i> (5), <i>andreevich</i> (8)	Rare transliterated names and low-frequency emotive vocabulary (GloVe artefact)
PC1	-		<i>any</i> (465), <i>government</i> (63), <i>than</i> (533), <i>its</i> (567), <i>people</i> (557), <i>us</i> (422)	High-frequency function words and abstract nouns (GloVe artefact)
PC2	+	1.855	<i>regiment</i> (226), <i>infantry</i> (82), <i>cavalry</i> (77), <i>battalion</i> (44), <i>squadron</i> (64), <i>brigade</i> (5)	Military units and formations (War)
PC2	-		<i>really</i> (203), <i>yourself</i> (117), <i>feel</i> (133), <i>myself</i> (148), <i>laugh</i> (60), <i>everybody</i> (118)	Personal and emotional discourse (Peace)
PC3	+	2.045	<i>walked</i> (70), <i>sat</i> (345), <i>feet</i> (112), <i>yards</i> (20), <i>screaming</i> (6), <i>clutching</i> (12)	Physical action and embodied presence
PC3	-		<i>principles</i> (7), <i>undertake</i> (10), <i>renounce</i> (8), <i>utmost</i> (12), <i>justification</i> (11), <i>obligations</i> (6)	Moral obligation and abstract reasoning

4.2.4. Ulysses

Ulysses contains $N = 264,165$ raw tokens and $M = 101,489$ retained tokens, with $\alpha = 0.763$ and $\alpha_0 = 0.80$. Its first ten components account for 25.0% of the variance, the smallest fraction among the four texts, consistent with the diversity of registers across its eighteen episodes.

PC1 is strongly affected by Dublin-specific proper names, multilingual fragments, and neologisms. PC2 separates formal titles and institutional naming from colloquial and intimate discourse. PC3 contrasts physical and material description with the vocabulary of kinship, marriage and literary identity. These semantic poles are listed in Table 6.

Taken together, the four texts show that PCA produces an ordered static geometry whose leading directions are often semantically interpretable, but whose first component can be contaminated by peripheral lexical clusters. The temporal question is separate: whether visits along these directions occur regularly or in concentrated textual episodes.

Table 4: Top-6 word types at the positive and negative poles of PC1, PC2, PC3 and PC6 for *On the Origin of Species* ($R = 100$, $F_{\min} = 5$), with frequencies in the filtered token sequence. PC4 and PC5 are omitted; PC6 is included because it carries the largest burstiness of this text. B_k is a property of the component, not of the pole, and is therefore shown once per pair; the largest value over the $K = 20$ components is shown in bold.

PC	Pole	B_k	Top words (freq)	Interpretation
PC1	+	1.255	<i>sessile</i> (11), <i>faunas</i> (14), <i>ovules</i> (6), <i>plumage</i> (9), <i>polymorphic</i> (9), <i>insectivorous</i> (6)	Specialised 19th-c. natural history lexicon
PC1	-		<i>he</i> (163), <i>him</i> (17), <i>said</i> (47), <i>us</i> (104), <i>she</i> (24), <i>his</i> (90)	Pronouns and interpersonal register (GloVe artefact)
PC2	+	1.888	<i>northern</i> (37), <i>islands</i> (131), <i>sea</i> (100), <i>southern</i> (43), <i>eastern</i> (13), <i>island</i> (60)	Places and biogeographical setting
PC2	-		<i>innate</i> (7), <i>slightest</i> (7), <i>understand</i> (74), <i>infer</i> (31), <i>absolutely</i> (25), <i>judgment</i> (12)	Scientific reasoning and theoretical inference
PC3	+	1.727	<i>me</i> (153), <i>eat</i> (5), <i>beaks</i> (10), <i>squirrels</i> (5), <i>legs</i> (37), <i>rabbit</i> (5)	Field observations, specific specimens, first person
PC3	-		<i>geographical</i> (42), <i>reduction</i> (5), <i>proportional</i> (16), <i>increased</i> (27), <i>increase</i> (66), <i>corresponding</i> (36)	Quantitative comparison and proportional reasoning
PC6	+	1.900	<i>seas</i> (13), <i>arctic</i> (38), <i>climate</i> (85), <i>warmer</i> (12), <i>deserts</i> (6), <i>shores</i> (24)	Climate and physical environment (glacial-period chapters)
PC6	-		<i>character</i> (136), <i>named</i> (12), <i>female</i> (24), <i>written</i> (7), <i>characters</i> (170), <i>original</i> (24)	Taxonomic description and nomenclature (partly register-driven)

Table 5: Top-6 word types at the positive and negative poles of PC1, PC2, and PC3 for *Great Expectations* ($R = 100$, $F_{\min} = 5$).

PC	Pole	B_k	Top words (freq)	Interpretation
PC1	+	1.228	<i>jaggers</i> (242), <i>townsman</i> (6), <i>stammered</i> (6), <i>peeped</i> (5), <i>biddy</i> (231), <i>provis</i> (70)	Character names and Victorian dialect (GloVe artefact)
PC1	-		<i>will</i> (174), <i>its</i> (151), <i>government</i> (5), <i>their</i> (175), <i>any</i> (268), <i>year</i> (25)	High-frequency generic and abstract vocabulary
PC2	+	1.209	<i>river</i> (59), <i>near</i> (66), <i>opened</i> (41), <i>fell</i> (55), <i>port</i> (7), <i>yards</i> (9)	Spatial and kinetic language (river, marshes, forge)
PC2	-		<i>really</i> (41), <i>understand</i> (53), <i>think</i> (241), <i>anybody</i> (30), <i>myself</i> (235), <i>yourself</i> (51)	Introspective and conversational register
PC3	+	1.244	<i>legs</i> (35), <i>walked</i> (45), <i>feet</i> (34), <i>crying</i> (15), <i>knees</i> (15), <i>staring</i> (26)	Bodily action and gesture
PC3	-		<i>concerning</i> (13), <i>expressly</i> (5), <i>regarding</i> (8), <i>relation</i> (12), <i>undertake</i> (5), <i>subordinate</i> (6)	Formal Latinate register

Table 6: Top-6 word types at the positive and negative poles of PC1, PC2 and PC3 for *Ulysses* ($R = 100$, $F_{\min} = 5$), with frequencies in the filtered token sequence. B_k is a property of the component, not of the pole, and is therefore shown once per pair; the largest value over the $K = 20$ components is shown in bold. A word type may appear at opposite poles of different components: the components are orthogonal directions, not a partition of the vocabulary.

PC	Pole	B_k	Top words (freq)	Interpretation
PC1	+	1.286	<i>wonderworker</i> (6), <i>dedalus</i> (174), <i>maggy</i> (9), <i>purefoy</i> (20), <i>dignam</i> (83), <i>voglio</i> (9)	Dublin proper names, neologisms and multilingual fragments (GloVe artefact)
PC1	−		<i>government</i> (14), <i>any</i> (191), <i>than</i> (169), <i>people</i> (78), <i>because</i> (219), <i>year</i> (71)	Generic and institutional English (GloVe artefact)
PC2	+	1.495	<i>st</i> (6), <i>william</i> (40), <i>edward</i> (17), <i>sir</i> (219), <i>attended</i> (7), <i>frederick</i> (10)	Formal titles and institutional naming
PC2	−		<i>yourself</i> (38), <i>really</i> (49), <i>feel</i> (104), <i>stupid</i> (11), <i>stuff</i> (24), <i>ourselves</i> (10)	Colloquial and intimate register
PC3	+	1.329	<i>thick</i> (26), <i>inches</i> (7), <i>surface</i> (10), <i>tail</i> (24), <i>mud</i> (6), <i>sloping</i> (9)	Physical and material description
PC3	−		<i>married</i> (56), <i>marry</i> (10), <i>poet</i> (29), <i>sir</i> (219), <i>son</i> (113), <i>brother</i> (66)	Kinship, marriage and literary identity

4.3. Burstiness across principal components

We now turn from static geometry to temporal organisation. For each component, threshold crossings define event sequences and IET distributions. Figure 3 and Figures 2, S1, S2 and S3 show the representative-text evidence. In every text, the original series has broader IET distributions and larger B_k than either control. The spectra are component dependent and generally largest among the early PCs, but the decay with component rank is not monotone. This non-monotonicity is expected because PCA orders variance rather than burstiness.

4.3.1. War and Peace

For PC1–PC3, the original IET distributions extend well beyond the geometric reference, whereas the shuffled and FGN controls closely follow it (Figure 2). The burstiness spectrum in Figure 3A is above both controls for all 20 components. PC3 has the largest value, $B_3 = 2.045$, approximately 2.1 times the geometric baseline and more than twice the corresponding null values ($B_3^{\text{sur}} = 0.976$, $B_3^{\text{FGN}} = 0.974$). Because events are threshold crossings of the upper tail, this measures the clustering of the positive pole of PC3: passages of physical action and embodied movement, which are concentrated in the battle and campaign chapters. The eigenvalue–burstiness correlation is positive but imperfect, $r = 0.702$ ($P = 0.0055$).

4.3.2. On the Origin of Species

The original projections have broad IET tails, particularly for PC2 and PC3, while both controls remain close to geometric (Figure S1). Burstiness is high

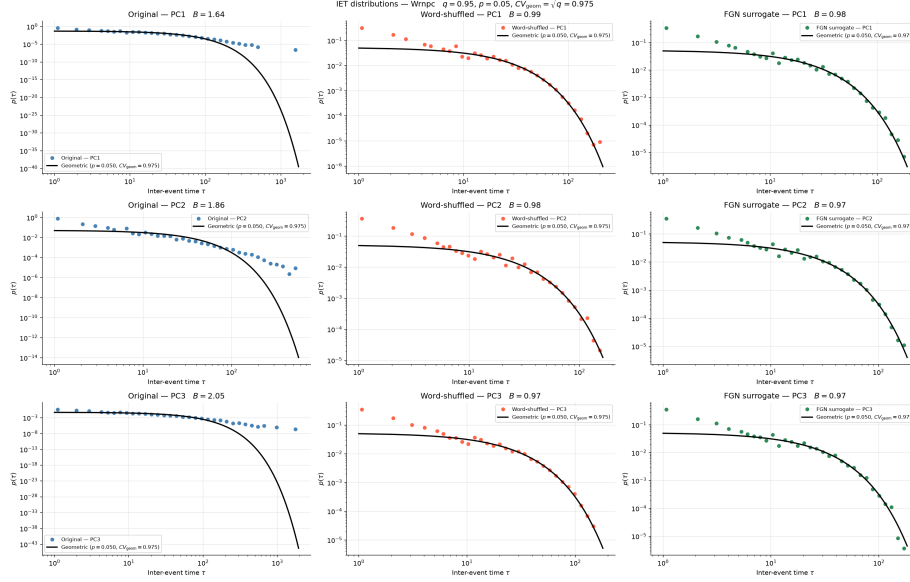


Figure 2: Inter-event-time distributions for PC1–PC3 of *War and Peace* at $q = 0.95$. Columns show the original text (blue), word-shuffled null (red), and FGN surrogate (green). The original projections have broad tails, whereas both controls lie close to the geometric reference.

across the leading directions but peaks away from the leading eigenvalue: the two largest values, $B_6 = 1.900$ and $B_2 = 1.888$, differ by less than one per cent and are followed by $B_3 = 1.727$, while PC1 is markedly lower at $B_1 = 1.255$. Both PC6 and PC2 have positive poles tied to specific parts of the book — climate and the glacial period, and places and biogeography respectively (Table 4) — which is what concentrates their threshold crossings. The full original-over-control ordering holds for all 20 components. The linear eigenvalue–burstiness correlation is weak, $r = 0.295$ ($P = 0.169$), while the rank correlation is moderate, $\rho = 0.460$ ($P = 0.042$): the association is monotone but not linear. That the burstiness maximum lies well away from PC1 states the point without recourse to any correlation measure — the direction of greatest semantic variance is not the direction of greatest temporal clustering.

4.3.3. Great Expectations

The original PC1–PC3 IET distributions again have broader tails than the controls (Figure S2). At $q = 0.95$, $B_1 = 1.228$, $B_2 = 1.209$ and $B_3 = 1.244$, while the null values lie in 0.95–1.01. The original remains above both controls over all 20 components (Figure 3C), but the spectrum is flatter and lower overall than for *War and Peace*: the largest value exceeds B_1 by only 1.3%, against 25% in Tolstoy, and the mean over the twenty components, 1.18, is the lowest of the four texts. Component-level burstiness is therefore present throughout the spectrum without being concentrated in any particular direction. The eigenvalue–

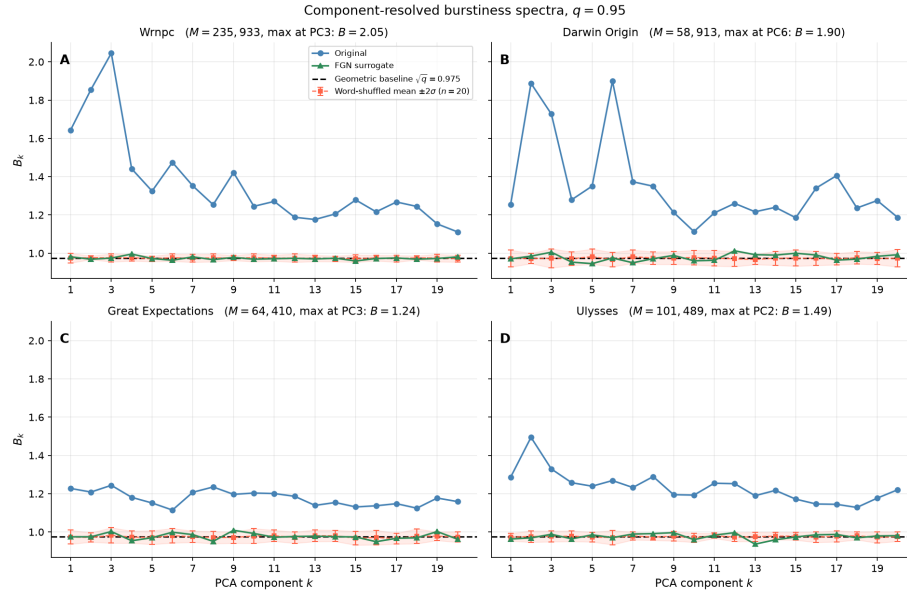


Figure 3: Component-resolved burstiness spectra at $q = 0.95$ for the four representative texts: **A** *War and Peace*, **B** *On the Origin of Species*, **C** *Great Expectations*, **D** *Ulysses*. In every panel the original (blue circles) lies above the FGN surrogate (green triangles), the word-shuffled mean $\pm 2\sigma$ band over 20 realisations (red), and the geometric baseline $CV_{\text{geom}} = \sqrt{q} \simeq 0.975$, for all 20 components. Events are crossings of the upper tail, so B_k measures the clustering of visits to the *positive* pole of each component (Tables 3–6). Burstiness is not monotone in PCA rank, because PCA orders components by variance rather than by temporal clustering: the maximum falls at PC3, PC6, PC3 and PC2 respectively, and never at PC1 — although in **B** and **C** the leading values are closely spaced.

burstiness correlation is $r = 0.511$ ($P = 0.014$), with a rank correlation $\rho = 0.579$ ($P = 0.009$).

4.3.4. Ulysses

For *Ulysses*, the original IET distributions are broad for all three displayed components, while both controls collapse towards the geometric reference (Figure S3). The spectrum peaks at PC2, $B_2 = 1.495$, with $B_1 = 1.286$ and $B_3 = 1.329$ (Figure 3D); the positive pole of PC2 is the register of formal titles and institutional naming (Table 6), which the novel deploys in concentrated parodic passages rather than evenly. The original exceeds both controls throughout the first 20 components. Here the two correlation measures diverge most strongly: $r = 0.571$ ($P = 0.041$) against $\rho = 0.817$ ($P < 10^{-4}$), so that the association between semantic variance and burstiness is close to monotone while being poorly described by a linear fit.

Averaged over the first 20 components at $q = 0.95$, the representative texts have $\bar{B}^{\text{orig}}$ of 1.36 (*War and Peace*), 1.35 (*On the Origin of Species*), 1.23 (*Ulysses*) and 1.18 (*Great Expectations*), against $\bar{B} \simeq 0.97$ – 0.98 for both null models in every case. The ordering is not a simple function of either text length or rank-sequence DFA exponent. The common feature is that component-level burstiness is present in the original semantic trajectories and absent from both null models.

4.4. Robustness to the threshold quantile and to the number of components

The component separation is not tied to the default $q = 0.95$. Figure S4 shows the mean burstiness over $k = 1, \dots, 20$ as q varies from 0.60 to 0.99; the corpus-level version of the ordering fraction is given in Section 5.3. For each of the four representative texts, the ordering holds for all 20 components throughout the displayed range. As expected, the absolute burstiness increases towards more extreme thresholds because rare visits to the upper tail accentuate episode-to-episode clustering.

The separation is likewise not an artefact of truncating the spectrum at $K = 20$. The retained components account for 35–38% of the embedding variance; refitting the PCA with $K = 50$ and recomputing the spectrum at $q = 0.95$ leaves every component above the independent-event baseline in all four texts, with a minimum of $B_k = 1.041$, some 7% above CV_{geom} . Burstiness does decline with component index: the mean falls from 1.18–1.36 over the first twenty components to 1.11–1.19 over components 21–50, while the variance carried per component falls from 1.9% to 0.65%. The choice $K = 20$ is conventional rather than principled; what the extension to $K = 50$ shows is that it does not act as a threshold beyond which component-level burstiness disappears. The leading components are also the ones for which the semantic poles were examined (Section 4.2).

Table 7: Corpus of 16 literary and scientific texts. N : total token count after tokenisation. M : content tokens after frequency filtering (top $R = 100$ words removed, minimum frequency 5). α : DFA scaling exponent of the rank sequence. α_0 : FGN generator exponent matched to α (bisection tolerance 0.01).

Title	Author	N	M	α	α_0
Pride and Prejudice	Austen	122,226	42,364	0.642	0.67
Don Quixote (transl.)	Cervantes	403,152	147,279	0.711	0.76
On the Origin of Species	Darwin	151,179	58,913	0.770	0.80
The Voyage of the Beagle	Darwin	206,728	84,931	0.629	0.67
David Copperfield	Dickens	363,557	134,409	0.727	0.78
Great Expectations	Dickens	188,925	64,410	0.689	0.73
Oliver Twist	Dickens	161,529	59,729	0.711	0.76
Ulysses	Joyce	264,165	101,489	0.763	0.80
The Jungle Book	Kipling	151,161	49,522	0.775	0.83
Moby Dick	Melville	215,652	82,773	0.694	0.73
Principia Mathematica	Newton	112,225	26,347	0.815	0.85
The Analysis of Mind	Russell	88,342	30,449	0.743	0.76
War and Peace	Tolstoy	564,788	235,933	0.721	0.76
Huckleberry Finn	Twain	145,924	52,741	0.774	0.83
Tom Sawyer	Twain	71,102	22,085	0.788	0.80
The Picture of Dorian Gray	Wilde	55,619	16,776	0.845	0.87

5. Corpus-level generalisation

Having established the component-resolved pattern in four representative texts (Section 4), we now assess its generality across the full corpus.

5.1. Corpus description

The corpus comprises 16 English literary texts from Project Gutenberg, spanning novels, scientific treatises and travel writing, in English editions and translations published between the eighteenth and early twentieth centuries. (Table 7). Raw token counts range from approximately 56,000 (The Picture of Dorian Gray) to 565,000 (War and Peace), with filtered token counts M (after high-frequency and hapax removal, OOV excluded) between 17,000 and 236,000. DFA scaling exponents α lie in the range $[0.63, 0.85]$, consistent with the literature on long-range correlations in English texts [4, 8].

Per-text IET distributions and burstiness spectra for all texts and threshold values can be regenerated with `scripts/run_text.py` in the code repository. The main text reports the aggregated corpus results.

5.2. Component-resolved burstiness holds corpus-wide

The corpus-level component spectrum is summarised in Figure 4. Averaging the burstiness spectrum $\{B_k\}$ across all 16 texts at quantile $q = 0.95$, the original texts are systematically more bursty than both null models across all

20 PCA components: $\langle B_k^{\text{orig}} \rangle_{\text{texts}} \in [1.16, 1.54]$ (mean 1.26), while both null models cluster tightly around the geometric baseline $\text{CV}_{\text{geom}} = \sqrt{0.95} \approx 0.975$: $\langle B_k^{\text{surr}} \rangle \in [0.973, 0.978]$ and $\langle B_k^{\text{FGN}} \rangle \in [0.958, 0.983]$.

At the level of individual (text, k) pairs ($16 \times 20 = 320$ pairs):

- $B_k^{\text{orig}} > B_k^{\text{surr}}$ in **100%** of pairs.
- $B_k^{\text{orig}} > B_k^{\text{FGN}}$ in **100%** of pairs.
- $|B_k^{\text{FGN}}/B_k^{\text{surr}} - 1| < 0.05$ in **95.6%** of pairs.

The appropriate summary statistic for the magnitude of the effect is the *per-text mean ratio*, computed by averaging $B_k^{\text{orig}}/B_k^{\text{surr}}$ over the 20 components for each text and then averaging over the 16 texts:

$$\left\langle \frac{B_k^{\text{orig}}}{B_k^{\text{surr}}} \right\rangle_{\text{texts}} = 1.29 \pm 0.03, \quad t = 8.7, \quad P < 10^{-6}, \quad (17)$$

where $\pm$ denotes the standard error of the mean over 16 texts. The original texts are thus on average $\sim 30\%$ more bursty than the word-shuffled null, which itself sits at the geometric baseline. The per-text values range from 1.17 (*Pride and Prejudice*) to 1.73 (*Principia Mathematica*), with a spread across texts (sd = 0.13) that is uncorrelated with text length ($r = -0.04$ against M). The four expository texts of the corpus occupy four of the five highest positions (mean 1.45 against 1.24 for the twelve narrative works), which is consistent with the chapter-level topical segmentation of expository prose; with four non-fiction texts this is an observation to be tested on a larger and deliberately balanced corpus, not a result. The analogous ratio for the FGN surrogate is $\langle B_k^{\text{orig}}/B_k^{\text{FGN}} \rangle = 1.29 \pm 0.03$, indistinguishable from the word-shuffled value.

The two null models satisfy $\langle B_k^{\text{FGN}}/B_k^{\text{surr}} \rangle = 1.001 \pm 0.001$ (range over texts $[0.996, 1.005]$; $t = 1.1$, $P = 0.28$): the FGN surrogate and the word-shuffled null are statistically indistinguishable at the corpus level (Section 5.4).

5.3. Robustness to the threshold quantile

Figure 5 shows the fraction of (text, k) pairs satisfying the ordering conditions as a function of $q \in \{0.60, 0.70, 0.80, 0.90, 0.95, 0.99\}$.

The full ordering $B_k^{\text{orig}} > B_k^{\text{surr}}$ and $B_k^{\text{orig}} > B_k^{\text{FGN}}$ is satisfied in 100% of pairs for all $q \in [0.60, 0.95]$, and in 98.1% of pairs at the extreme $q = 0.99$. The null-model indistinguishability criterion $|B_k^{\text{FGN}}/B_k^{\text{surr}} - 1| < 0.05$ is met by $\geq 99\%$ of pairs for $q \leq 0.90$, by 96% at $q = 0.95$, and decreases to 73% at $q = 0.99$.

This degradation at the extreme threshold is expected: at the corpus median, $M \approx 59,000$, the event rate $p = 0.01$ leaves only ~ 590 events per series, giving a relative error on the IET estimator of $\sim 4\%$, rising to $\sim 8\%$ for the shortest text ($M = 16,776$) — comparable to or larger than the magnitude of $B^{\text{FGN}} - B^{\text{surr}}$ at this threshold. For $q = 0.95$ the same texts give $\sim 3,000$ and ~ 840 events, with

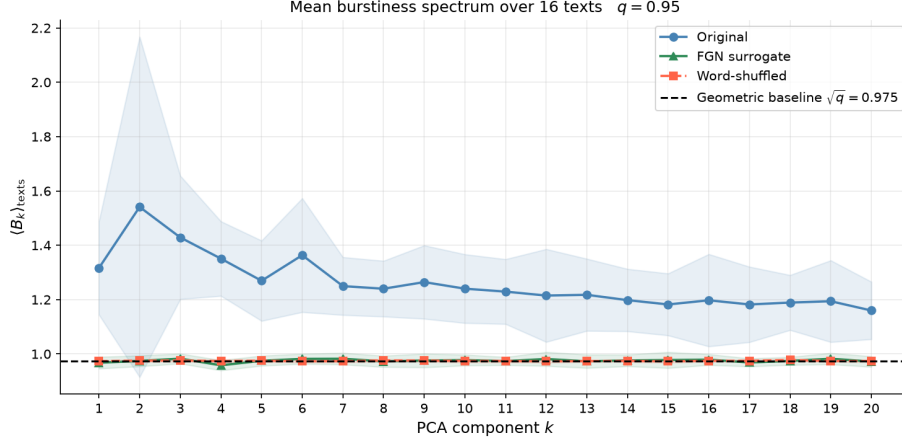


Figure 4: Mean component-resolved burstiness spectrum across the analysed corpus at $q = 0.95$. Curves show the mean $\langle B_k \rangle$ over texts for $k = 1, \dots, 20$; shaded bands indicate one standard deviation. The original texts (blue) are above the FGN surrogate (green), word-shuffled null (red), and geometric baseline $\sqrt{q} \simeq 0.975$ at every component. The largest corpus means occur among the early components, followed by a broad decline with non-monotone secondary peaks.

relative errors of 1.8% and 3.5%, well within the regime where the null-model comparison is reliable.

The same estimator noise accounts for the six pairs out of 320 that fail the full ordering at $q = 0.99$.

5.4. Distribution of burstiness ratios

Figure S5 shows the distribution of the ratios $B^{\text{orig}}/B^{\text{surr}}$, $B^{\text{orig}}/B^{\text{FGN}}$, and $B^{\text{FGN}}/B^{\text{surr}}$ over all 320 (text, k) pairs at $q = 0.95$.

Panel A shows that both orig–null ratios are concentrate well above 1, with no mass below 1. The two distributions (orig/surr and orig/FGN) are nearly identical in shape and location, confirming that the FGN and word-shuffled null models are equally poor surrogates for the original burstiness.

Panel B isolates the ratio $B^{\text{FGN}}/B^{\text{surr}}$, which is tightly concentrated around 1 over the 320 pairs (mean 1.0008, sd 0.022, range [0.91, 1.09]). Because components within a text are not independent, the test is performed on the 16 per-text means: a one-sample t -test against unity gives $t = 1.13$, $P = 0.28$. The two null models are statistically indistinguishable, and the distribution is symmetric with no indication of a directional bias.

Together, Figures 4–S5 establish the corpus-level result:

$$B_k^{\text{orig}} \gg B_k^{\text{FGN}} \approx B_k^{\text{surr}} \quad \forall k, \forall \text{text in corpus}, \quad (18)$$

with a mean excess of the original over the word-shuffled null of $\sim 30\%$. The ordering holds for every text and every component irrespective of length, genre, DFA exponent and choice of quantile threshold; its magnitude, by contrast, varies systematically between texts (Section 5.2).

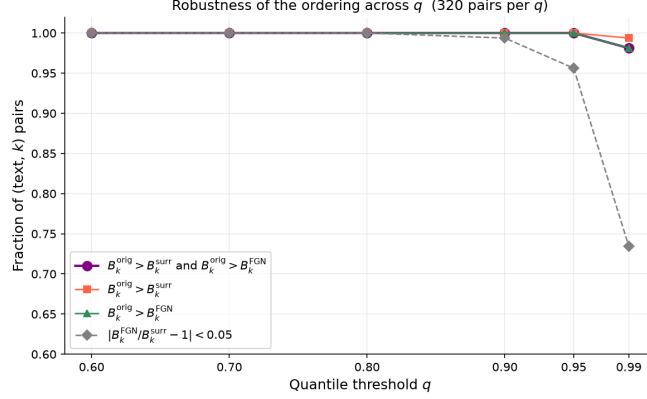


Figure 5: **Robustness of the burstiness ordering across quantile threshold q .** Each point: fraction of (text, k) pairs (320 per q) satisfying the indicated condition. Purple circles: full ordering $B_k^{\text{orig}} > B_k^{\text{surr}}$ and $B_k^{\text{orig}} > B_k^{\text{FGN}}$. Red squares: $B_k^{\text{orig}} > B_k^{\text{surr}}$. Green triangles: $B_k^{\text{orig}} > B_k^{\text{FGN}}$. Grey diamonds: $|B_k^{\text{FGN}}/B_k^{\text{surr}} - 1| < 0.05$. The ordering is robust for $q \in [0.60, 0.95]$; the slight degradation at $q = 0.99$ is due to estimator noise at $p = 0.01$.

6. Temporal persistence in semantic projection space

Sections 4 and 5 establish that the original semantic trajectories have elevated component-resolved burstiness and are separated from both controls. We now test the same temporal organisation in the continuous projections using normalised autocorrelation functions and DFA. An information-theoretic argument explains why the long-range correlations imposed on the FGN rank sequence need not survive the mapping into embedding space.

6.1. Conditional entropy of the embedding trajectory

[TODO: Mirko: drop it or develop it ! See % comments in tex file....]

The qualitative argument of Remark 3 can be cast in information-theoretic terms. Consider the embedding trajectory $\mathcal{E} = (\mathbf{e}_t)_{t=1}^M$ as a stationary (or locally stationary) stochastic process over $\mathbb{R}^d$. For a lag $\ell \geq 1$, define the *conditional entropy*

$$H_\ell = H(\mathbf{e}_t \mid \mathbf{e}_t[t-1], \dots, \mathbf{e}_t[t-\ell]), \quad (19)$$

where $\mathbf{e}_t[s] \equiv \mathbf{e}_s$ for brevity. For an i.i.d. process, $H_\ell = H(\mathbf{e}_t)$ for all $\ell \geq 1$: past embeddings carry no information about the present. For a process with long-range semantic structure (real text), $H_\ell < H(\mathbf{e}_t)$ and the gap $H(\mathbf{e}_t) - H_\ell$ grows with ℓ as long as topical context persists.

The corresponding working hypothesis is:

$$H_\ell^{\text{orig}} < H_\ell^{\text{FGN}} \approx H_\ell^{\text{shuf}} \approx H(\mathbf{e}_t), \quad \ell = 1, 2, \dots \quad (20)$$

The word-shuffled null is exchangeable and approaches the i.i.d. reference apart from finite-size dependence associated with fixed token counts; accordingly, $H_\ell^{\text{shuf}} \simeq H(\mathbf{e}_t)$. For the FGN surrogate, past ranks $r_{t-1}, \dots, r_{t-\ell}$ carry information about the present rank r_t (by the long-range memory of the FGN process); but the mapping $r \mapsto \phi(w_r)$ is semantically diffuse: words of similar Zipf rank span the full range of semantic content and are spread across all regions of the embedding space. Therefore past ranks carry negligible information about the present embedding $\mathbf{e}_t$, and $H_\ell^{\text{FGN}} \approx H(\mathbf{e}_t)$. For the original text, past embeddings directly encode the topical context of the narrative, providing genuine predictive information about subsequent embeddings.

6.2. A proxy measure: autocorrelation of projections

Direct estimation of H_ℓ in $\mathbb{R}^{300}$ requires large samples and density estimation in high dimensions, which is impractical for individual texts of length $M \sim 10^4$ – 10^5 . We therefore use a computationally tractable proxy: the normalised autocorrelation function (ACF) of the projected scalar series,

$$\rho_k(\ell) = \frac{\text{Cov}(x_t^{(k)}, x_{t+\ell}^{(k)})}{\text{Var}(x_t^{(k)})}, \quad (21)$$

for PCA component k and lag ℓ . For an i.i.d. process (word-shuffled null), $\rho_k(\ell) = 0$ for all $\ell \geq 1$ by construction. For the original text, $\rho_k(\ell) > 0$ with slow decay, reflecting the long-range topical structure. For the FGN surrogate, the entropy argument of Section 6.1 predicts $\rho_k(\ell) \approx 0$ despite the long-range correlations in the rank sequence.

Remark 4. *The ACF proxy is connected to the burstiness spectrum through the Wiener–Khinchine theorem: a slowly decaying $\rho_k(\ell)$ corresponds to excess power at low frequencies in the spectral density of $x_t^{(k)}$. For a Gaussian process the level-crossing statistics are determined by the autocorrelation; in general the two are related but not equivalent, since B_k depends on the full IET distribution while ρ_k is a second-order statistic. In the present data they move together: the original has both elevated B_k and slowly decaying ρ_k , and both null models have neither.*

6.3. Empirical results: ACF of projections

Figure 6 shows $\rho_k(\ell)$ as a function of lag ℓ (log scale) for PC1, PC2, and PC3 of *Great Expectations* by Dickens ($\alpha = 0.689$), for the original text, the FGN surrogate, and the word-shuffled null. Figure S6 shows the same for the mean ACF over all $K = 20$ components.

Original text. The ACF is positive and significantly above the i.i.d. band $\pm 1.96/\sqrt{M} \approx \pm 0.008$ at lag $\ell = 1$ for all 20 components, with $\rho_k(1) \in [0.049, 0.208]$ (Table S1). The decay is gradual on a log scale, with values remaining above the i.i.d. band up to lags of order $\ell \sim 100$ for the lower-indexed components. This slow decay is a second-order signature of the temporal persistence that accompanies semantic burstiness.

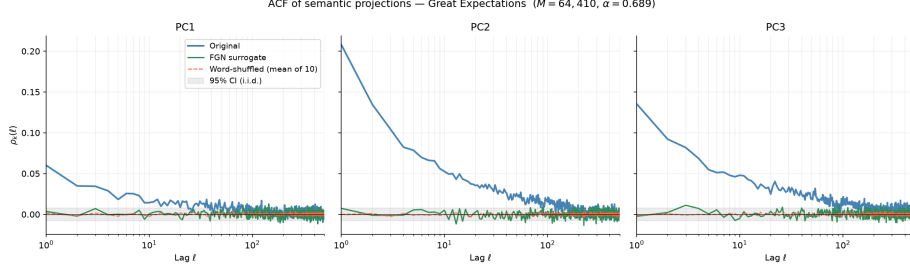


Figure 6: **Autocorrelation function $\rho_k(\ell)$ of projected series, as a function of lag ℓ (log scale) — *Great Expectations* (Dickens).** Each panel shows one PCA component (PC1, PC2, PC3). Blue: original filtered text. Green: FGN surrogate. Red dashed: word-shuffled null. Grey band: $\pm 1.96/\sqrt{M}$ i.i.d. reference (95% CI). The original text has substantial positive ACF that decays slowly on a log scale, remaining above the i.i.d. band up to $\ell \sim 100$. Both null models collapse immediately into the i.i.d. band, despite the FGN surrogate having the same DFA exponent $\alpha = 0.689$ as the original.

FGN surrogate. All 20 components have $|\rho_k^{\text{FGN}}(1)| \leq 0.007$, inside the i.i.d. band. At lag $\ell = 100$ all values are inside the band. The FGN surrogate, despite matching the DFA exponent of the original ($\alpha = 0.689$), exhibits only negligible autocorrelation in the projected series: rank-level memory does not reach the thematic directions of semantic space.

Word-shuffled null. As expected, $\rho_k^{\text{surr}}(\ell) \approx 0$ for all k and ℓ , consistent with the random-permutation reference.

6.4. Corpus-level ACF results

We computed $\rho_k(\ell)$ for all 16 texts and all $K = 20$ PCA components at lags $\ell \in \{1, 2, 5, 10, 20, 50, 100, 200, 300\}$, yielding $16 \times 20 \times 9 = 2,880$ measurements. The 95% confidence band for an i.i.d. process at sample size M is $\pm 1.96/\sqrt{M}$; since M varies across texts, we use the text-specific CI for significance tests.

Mean ACF decay. Figure S7 shows the mean $\rho(\ell)$ averaged over all (text, k) pairs as a function of lag. The original text maintains positive and significant autocorrelation across the full range of lags tested, with mean values

$$\bar{\rho}^{\text{orig}}(1) = 0.100 \pm 0.045, \quad \bar{\rho}^{\text{orig}}(10) = 0.032 \pm 0.021, \quad \bar{\rho}^{\text{orig}}(100) = 0.011 \pm 0.011, \quad (22)$$

where $\pm$ denotes one standard deviation over (text, k) pairs. The FGN surrogate and the word-shuffled null are indistinguishable from each other and from zero across all lags:

$$\bar{\rho}^{\text{FGN}}(1) = 0.002 \pm 0.005, \quad \bar{\rho}^{\text{surr}}(1) = 0.000 \pm 0.004. \quad (23)$$

The ratio $\bar{\rho}^{\text{orig}}(1)/\bar{\rho}^{\text{FGN}}(1) \approx 50$ quantifies how completely the FGN rank-to-embedding mapping destroys the frequency-domain long-range correlations in semantic projection space.

Fraction of significant pairs. Figure S8 shows, for each lag, the fraction of (text, k) pairs where $|\rho_k(\ell)|$ exceeds the i.i.d. band. At lag $\ell = 1$: 100% of original-text pairs are significant, 10.9% of FGN pairs, and 4.7% of shuffled pairs. The FGN figure exceeds the expected false-positive rate of 5%, but the significant cases are concentrated on PC1 (56% of the PC1 pairs, against 8% for $k \geq 4$) — the component most affected by the geometric artefact of rare transliterated proper nouns in the GloVe embedding space (Section 3.1). From lag $\ell = 2$ onwards the FGN fraction lies between 2.5% and 7%, at the false-positive level. The effect is in any case tiny relative to the separation of the original projections from both controls.

The “original only” curve (orig significant, FGN not) peaks at 93% at lag $\ell = 2$ and remains above 50% up to lag $\ell = 100$, providing a clean corpus-level version of the single-text result. The original text maintains detectable semantic autocorrelation at scales of order 100 filtered tokens, corresponding to roughly 300 tokens of running text — a few paragraphs.

Distribution of $\rho_k(1)$. Figure S9 shows the distribution of $\rho_k(1)$ over all 320 (text, k) pairs. Panel A shows two completely separated distributions: the original has $\rho_k(1) \in [0.026, 0.264]$ with mean 0.100, all pairs significant; the FGN has $\rho_k(1) \in [-0.014, 0.023]$ with mean 0.002, the vast majority inside the i.i.d. band.

Panel B zooms on the FGN and shuffled distributions. Both are centred near zero and overlap substantially. A one-sample t -test of $\bar{\rho}^{\text{FGN}}(1) = 0$ gives $t = 5.05$, $P < 10^{-6}$: the mean is statistically non-zero but negligibly small (0.002) relative to the orig–null gap (0.098). As with the burstiness differences (Section 5.4), the statistical significance at this sample size reflects estimator sensitivity rather than a scientifically meaningful effect.

Remark 5. *The residual FGN autocorrelation on PC1 is expected rather than anomalous. Rank is an ordering by frequency, so the long-range memory imposed on $\mathbf{r}$ is memory along whichever direction of embedding space encodes frequency — and in every text of the corpus that direction is PC1, whose poles oppose rare proper nouns and transliterations to generic high-frequency vocabulary (Section 4.2). The observation that 56% of PC1 pairs but only 8% of pairs with $k \geq 4$ show significant lag-one autocorrelation is therefore the signature of rank-level memory reaching the one direction it can reach, and failing to reach the thematic ones.*

6.5. DFA of projected series

The ACF results of Sections 6.3 and 6.4 confirm the predicted ordering $\rho_k^{\text{orig}}(\ell) \gg \rho_k^{\text{FGN}}(\ell) \approx \rho_k^{\text{surr}}(\ell) \approx 0$. We now complement the ACF analysis by computing the DFA scaling exponent of the projected series $x_t^{(k)}$ directly, providing the semantic-level analogue of the symbolic-level DFA result that motivates the paper.

For each PCA component k and each of the three series (original, FGN, word-shuffled), we apply DFA to the scalar time series $x_t^{(k)}$ and estimate the

scaling exponent α_k from $F(s) \sim s^{\alpha_k}$, where $F(s)$ is the detrended fluctuation function at box size s . Box sizes range from $s_{\min} = 10$ to $s_{\max} = M/4$ on a logarithmic grid.

Results on Great Expectations. Table S2 and Figure 7 report α_k for all $K = 20$ components of *Great Expectations* by Dickens ($\alpha = 0.689$). The results confirm the prediction precisely:

- **Original:** $\alpha_k^{\text{orig}} > 0.5$ in all 20/20 components, with mean $\bar{\alpha}^{\text{orig}} = 0.623 \pm 0.034$. All values exceed the shuffled reference by more than 2σ , confirming long-range correlations in every semantic projection.
- **FGN surrogate:** $|\alpha_k^{\text{FGN}} - 0.5| < 0.05$ in all 20/20 components, with mean $\bar{\alpha}^{\text{FGN}} = 0.499 \pm 0.014$. Despite having a matched DFA exponent ($\alpha = 0.689$, generator exponent $\alpha_0 = 0.74$) in the rank sequence by construction, the FGN projects onto every semantic direction as an uncorrelated process.
- **Word-shuffled:** $\bar{\alpha}^{\text{surr}} = 0.499 \pm 0.004$, consistent with $\alpha = 0.5$ (white noise) as expected for an i.i.d. sequence.

Figure 8 shows the log-log fluctuation function $F(s)$ for PC1, PC2, and PC3. The original text shows a clear power-law scaling with slope $\alpha_k \approx 0.60$ – 0.69 , visibly steeper than the reference slope 0.5. The FGN and shuffled curves are parallel to the reference slope, confirming $\alpha_k \approx 0.5$ for both null models.

Remark 6. *The exponents $\alpha_k^{\text{orig}} \in [0.564, 0.693]$ do not decrease monotonically with k : PC12 ($\alpha = 0.642$) and PC15 ($\alpha = 0.645$) exceed PC1 ($\alpha = 0.605$), even though they carry an order of magnitude less variance. This partial decorrelation between PCA rank (ordered by variance) and DFA exponent mirrors the decorrelation between the eigenvalue spectrum $\{\lambda_k\}$ and the burstiness spectrum $\{B_k\}$ reported in Section 4: semantic variance and semantic temporal structure are distinct properties of narrative organisation.*

Consistent temporal signatures. The DFA, ACF, and burstiness analyses provide mutually consistent but non-equivalent summaries of temporal organisation. For the original projections, $\alpha_k > 0.5$, the ACF remains positive over extended lags, and B_k exceeds the geometric reference. For both controls, $\alpha_k \simeq 0.5$, the ACF is close to zero, and $B_k \simeq \text{CV}_{\text{geom}}$. These observations can be summarised as

$$\alpha_k^{\text{orig}} > 0.5, \quad \rho_k^{\text{orig}}(\ell) > 0 \text{ over extended lags}, \quad B_k^{\text{orig}} > \text{CV}_{\text{geom}}, \quad (24)$$

whereas the FGN and shuffled projections remain close to their uncorrelated references in all three measures.

The FGN surrogate is especially informative because its rank sequence has the prescribed long-range DFA exponent, while every analysed semantic projection has $\alpha_k^{\text{FGN}} \simeq 0.5$. Rank-level persistence therefore does not propagate automatically into semantic projection space.

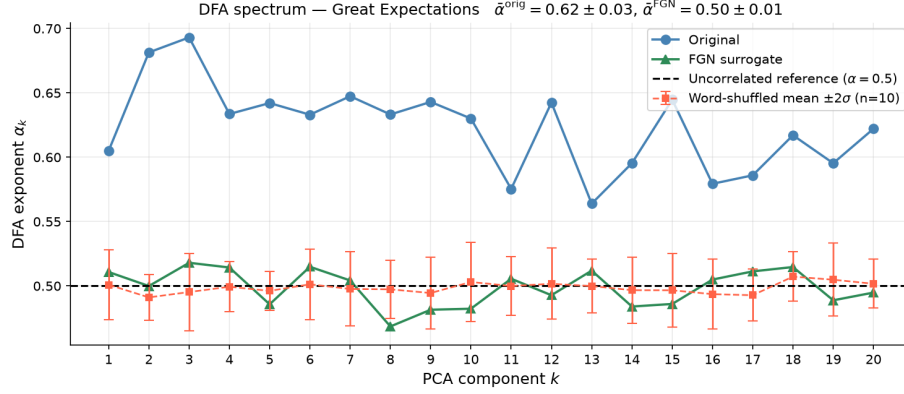


Figure 7: **DFA scaling exponent α_k of projected series $x_t^{(k)}$ as a function of PCA component k — *Great Expectations* (Dickens).** Blue circles: original filtered text. Green triangles: FGN surrogate. Red squares: word-shuffled null (mean $\pm 2\sigma$ over 10 realisations). Dashed line: uncorrelated reference $\alpha = 0.5$. The original text has $\alpha_k > 0.5$ for all 20 components; the FGN and shuffled are indistinguishable from $\alpha = 0.5$.

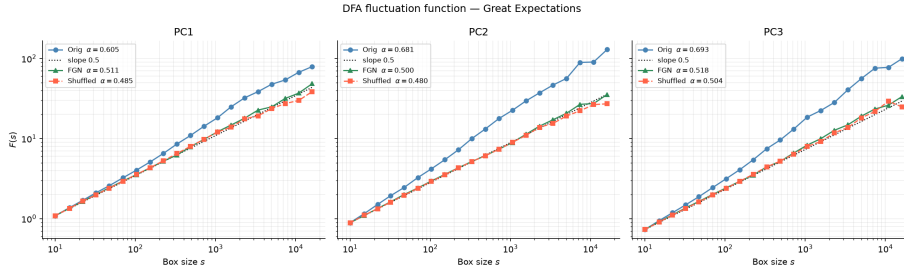


Figure 8: **Log-log DFA fluctuation function $F(s)$ for PC1, PC2, and PC3 — *Great Expectations* (Dickens).** Each panel shows one component; the estimated exponent α_k is given in the legend. Dotted line: reference slope 0.5. The original text (blue) has a consistently steeper slope than both null models, which are parallel to the $\alpha = 0.5$ reference across all box sizes s .

6.6. Connection between ACF and burstiness

The ACF and the burstiness parameter probe different aspects of the same projected series. By the Wiener–Khinchine theorem, slow ACF decay is associated with excess low-frequency power in the spectral density

$$S_k(\omega) = \sum_{\ell=-\infty}^{\infty} \rho_k(\ell) e^{-i\omega\ell}. \quad (25)$$

Such low-frequency variation is consistent with extended semantic episodes and clustered threshold exceedances, and therefore with broad IET distributions; in the long-range correlated case the sum in (25) diverges as $\omega \rightarrow 0$, which is the spectral expression of that persistence. The connection is not one-to-one, however: thresholding is non-linear, and B_k depends on temporal structure beyond the second-order information contained in the ACF.

The empirical conclusion is consequently based on the agreement of three complementary observations rather than on a deterministic causal identity. Original texts show elevated B_k , positive projection ACFs, and projection DFA exponents above 0.5. The FGN and word-shuffled controls remain close to the geometric baseline in burstiness, to zero in autocorrelation, and to white noise in DFA. This convergence of measures supports the interpretation that ordered topical organisation, rather than global rank-level scaling alone, produces persistence in semantic projection space.

7. Discussion and Conclusions

CRedit authorship contribution statement

Mirko Degli Esposti: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft, Writing – review & editing. **Marcelo A. Montemurro:** Conceptualization, Methodology, Formal analysis, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI in the writing process

During the preparation of this work the authors used a large language model to assist with code refactoring and manuscript editing. The authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Data availability

All texts are public-domain works from Project Gutenberg. The corpus, the analysis code, the result tables and the scripts that regenerate every figure in this paper are available at <https://github.com/mirko-degli-esposti/semantic-burstiness-texts>. Pre-trained GloVe vectors are distributed by Stanford NLP [10].

Acknowledgements

Appendix A. Geometric distribution as IET null: derivation

Consider a binary sequence $b_1, b_2, \dots, b_M$ where each b_t is independently Bernoulli(p). An *event* occurs at position t when $b_t = 1$. The inter-event time τ between two successive events is the number of positions (including the terminal event) from one event to the next. The probability of $\tau = k$ requires $b_{t+1} = \dots = b_{t+k-1} = 0$ and $b_{t+k} = 1$:

$$\Pr(\tau = k) = (1 - p)^{k-1} p, \quad k = 1, 2, 3, \dots \quad (\text{A.1})$$

which is the geometric distribution with parameter p . Its mean, variance, and coefficient of variation are

$$\langle \tau \rangle = \frac{1}{p}, \quad \text{Var}(\tau) = \frac{1 - p}{p^2}, \quad \text{CV}_{\text{geom}}(p) = \sqrt{1 - p}. \quad (\text{A.2})$$

The exponential distribution $p(\tau) \propto e^{-\lambda\tau}$ arises in the limit $p \rightarrow 0$ with $\lambda = -\log(1 - p) \approx p$, and has $\text{CV} = 1$ (Poisson process). For $p = 0.05$, $\text{CV}_{\text{geom}} = 0.975 \approx 1$, so the Poisson approximation is adequate. For $p = 0.5$, $\text{CV}_{\text{geom}} = 0.707$, and the Poisson approximation fails badly.

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